H **IATIONAL JUNIOR COLLEGE**

ENIOR HIGH 2 PRELIMINARY EXAMINATION

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CANDIDATE NAME			
SUBJECT CLASS		REGISTRATION NUMBER	

PHYSICS **9478/03**

Paper 3 Structured Questions

25 Aug 2026

Section A Booklet **2 hours**

Candidate answers on the Question Paper.

No Additional Materials are required.

READ THE INSTRUCTION FIRST

Write your subject class, registration number and name in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

Section A
Answers **all** questions.

You are advised to spend one and a half hours on Section A and half an hour on Section B

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**/ 7**
2	**/ 8**
3	**/ 8**
4	**/ 8**
5	**/ 10**
6	**/ 7**
7	**/ 7**
deduction	
Total (55)	

[Turn over

This document contains **18** printed pages and **2** blank pages.

DATA AND FORMULAE

Data

speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
	$(= 8.99 \times 10^9 \text{ m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\,\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\text{K} = T/^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

[Turn over

displacement of particle in s.h.m. $x = x_0 \sin \omega t$

velocity of particle in s.h.m. $v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$

electric current $I = nAvq$

resistors in series $R = R_1 + R_2 + \ldots$

resistors in parallel $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

capacitors in series $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$

capacitors in parallel $C = C_1 + C_2 + \ldots$

energy in a capacitor $U = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$

charging a capacitor $Q = Q_0 \left[1 - e^{-\frac{t}{\tau}}\right]$

discharging a capacitor $Q = Q_0 e^{-\frac{t}{\tau}}$

RC time constant $\tau = RC$

electric potential $V = \frac{Q}{4\pi\varepsilon_0 r}$

alternating current / voltage $x = x_0 \sin \omega t$

magnetic flux density due to a long straight wire $B = \frac{\mu_0 I}{2\pi d}$

magnetic flux density due to a flat circular coil $B = \frac{\mu_0 NI}{2r}$

magnetic flux density due to a long solenoid $B = \mu_0 nI$

energy states for quantum particle in a box $E_n = \frac{h^2}{8mL^2} n^2$

radioactive decay $x = x_0 e^{-\lambda t}$

radioactive decay constant $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in the spaces provided.

1 An elastic rope is attached to a 70.0 kg man on one end and to a bridge on the other end. The rope has a natural length of 25.0 m and an elastic constant of 120 N m^{-1}.

The man steps off the bridge and falls vertically downwards from rest.

Assume that air resistance acting on the man is negligible.

(a) Explain why the person has maximum speed when the tension in the elastic rope is equal to his weight.

……

……

……

…… [2]

(b) Determine the maximum speed of the man after he steps off the bridge.

maximum speed = …… m s^{-1} [3]

(c) Calculate the extension of the elastic rope when the man is at the lowest point of his motion.

extension = …….. m [2]

[Total: 7]

2 Fig. 2.1 shows a slope at 30° to the horizontal. P is a point on a slope and Q is the point the slope meets the horizontal ground. PQ is 1.2 m.

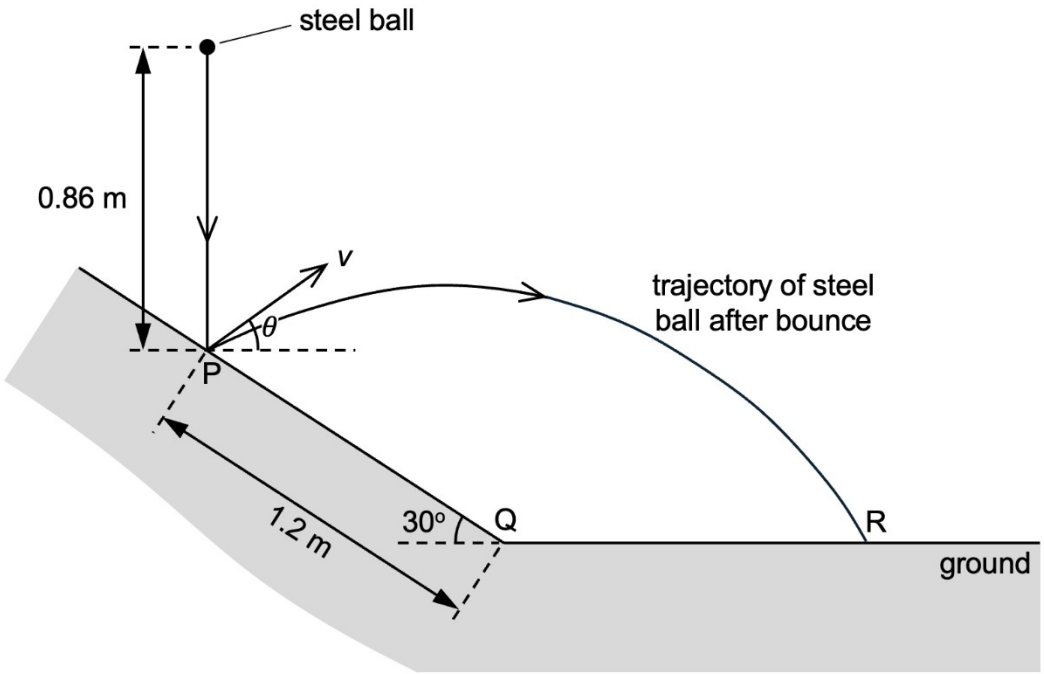


Fig. 2.1

A small steel ball is dropped from a height of 0.86 m above P.

The steel ball collides with the slope at P and bounces off elastically.

(a) Show that the speed of the steel ball just before the collision at P is 4.1 m s^{-1}.

[1]

(b) State the magnitude of the ball's velocity *v* immediately after the collision.

…… [1]

(c) The velocity *v* is at an angle θ to the horizontal as shown in Fig. 2.1.

Explain why $\theta = 30°$.

……

……

……

……

…… [3]

(d) Determine the distance QR.

QR = …….. m [3]

[Total: 8]

3 Fig 3.1 shows an open cube floating in a liquid.

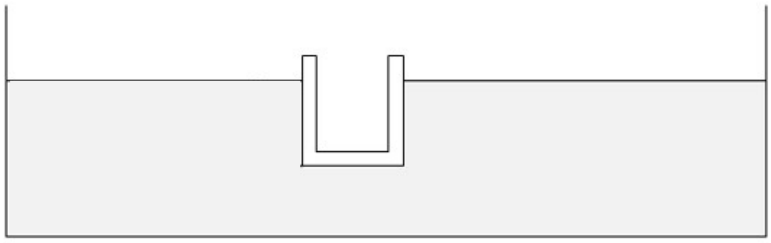


Fig. 3.1

(a) When the cube is displaced downwards by a small distance from the equilibrium position and released. The variation with time *t* of the vertical displacement *y* of the plate is shown in Fig. 3.2.

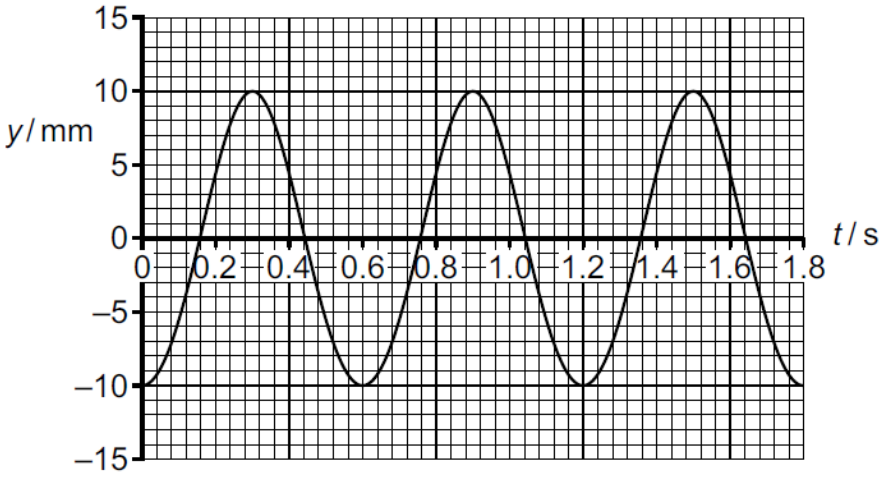


Fig. 3.2

The cube undergoes simple harmonic motion.

The mass *m* of the oscillating cube is 340 g.

(i) Determine the frequency of oscillation.

frequency =..... Hz [1]

(ii) Calculate the total energy of the oscillations.

energy =..... J [3]

[Turn over

(b) In an experiment, surface water waves are generated using a dipper shown in Fig. 3.3. The generated waves are incident on the cube, and produce vertical vibrations of constant amplitude but with a frequency that can be varied.

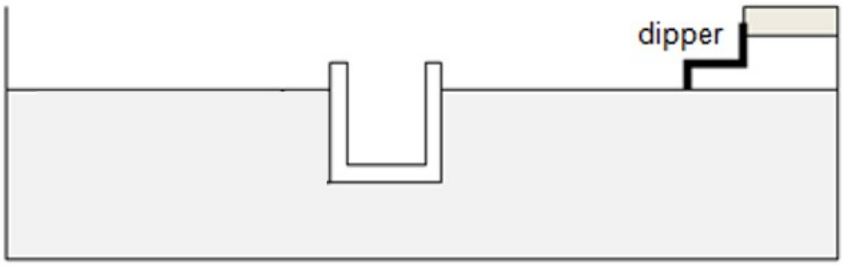


Fig. 3.3

The variation with frequency f of the amplitude y_0 of the oscillations of the cube is shown in Fig. 3.4.

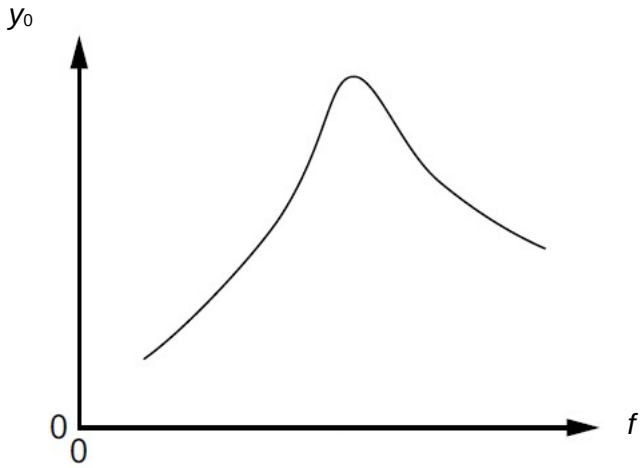


Fig. 3.4

(i) State the name of the phenomenon illustrated in Fig. 3.4.

……..

[1]

(ii) State the frequency at which y_0 is at its maximum.

frequency =..... Hz [1]

(iii) Sketch on Fig. 3.4 the effect on the oscillations if the liquid is more viscous. [2]

[Total: 8]

4 Two small speakers, L_1 and L_2, are separated by 15 cm, as shown in Fig. 4.1. A microphone M is moved along line RS parallel to the line joining the two speakers and at a perpendicular distance of 1.5 m away. P is a position on the line RS and equidistant from both speakers.

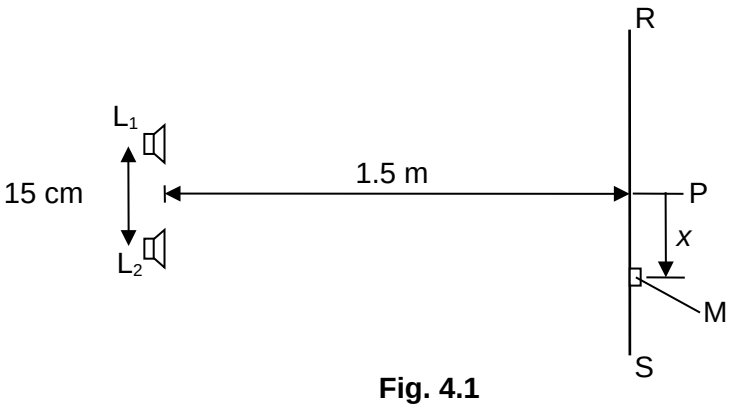


Fig. 4.1

The sound intensity detected by the microphone M varies with displacement *x* as M is moved from R towards S as shown in Fig. 4.2.

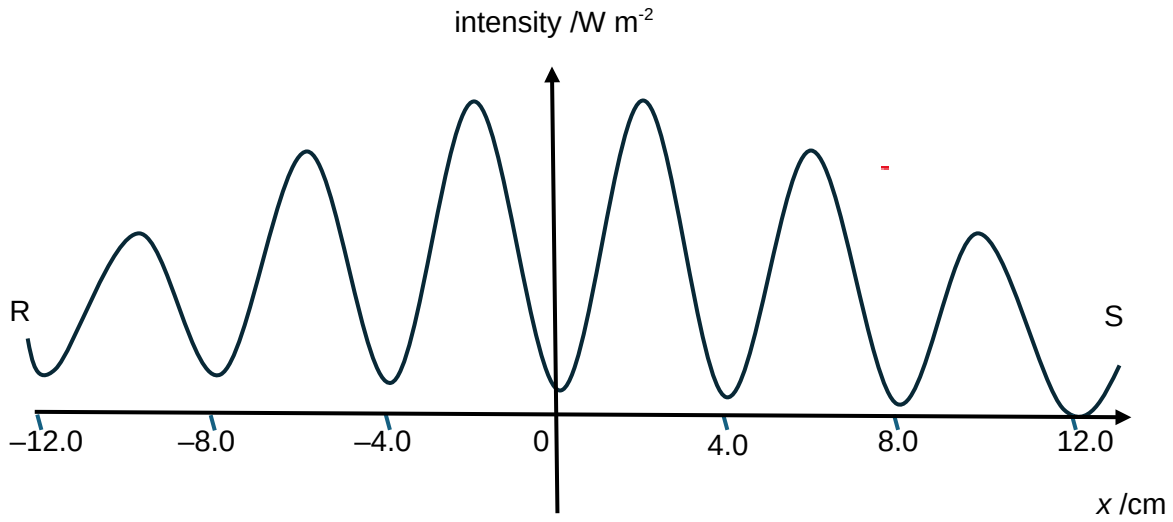


Fig. 4.2

(a) Explain how it can be deduced that the waves emitted by the two speakers are coherent.

……

…… [1]

(b) Explain, with reference to Fig. 4.2, how it can be deduced that the sound waves emitted by the speakers are not in phase.

……

……

……

……

……

…… [3]

(c) The speed of sound is 340 m s^{-1}.

Calculate the frequency of the sound waves emitted by the speakers.

frequency = …… Hz [2]

(d) Explain which speaker has a higher power.

……

……

……

…… [2]

[Total: 8]

5 Two uniformly charged non-conducting spheres A and B, of diameter 10 cm and 20 cm respectively, are situated in a vacuum. Their centres are separated by a distance of 50 cm, as shown in Fig. 5.1. *x* is the distance from the centre of sphere A to any arbitrary point along the line joining the centres of the two spheres.

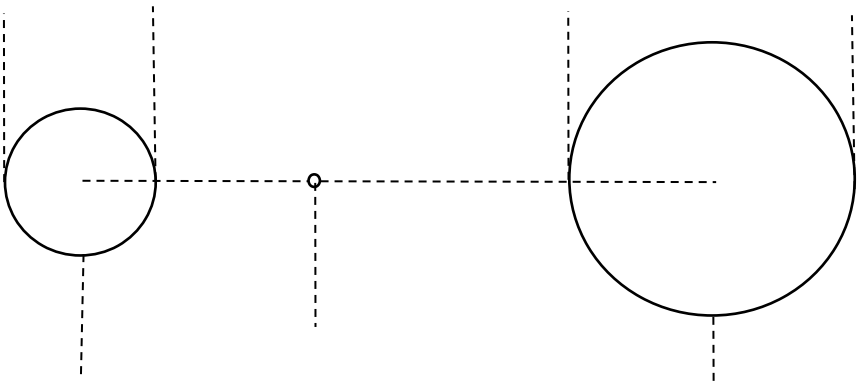


Fig. 5.1

The variation with *x* of the electric potential *V* is shown in Fig. 5.2.

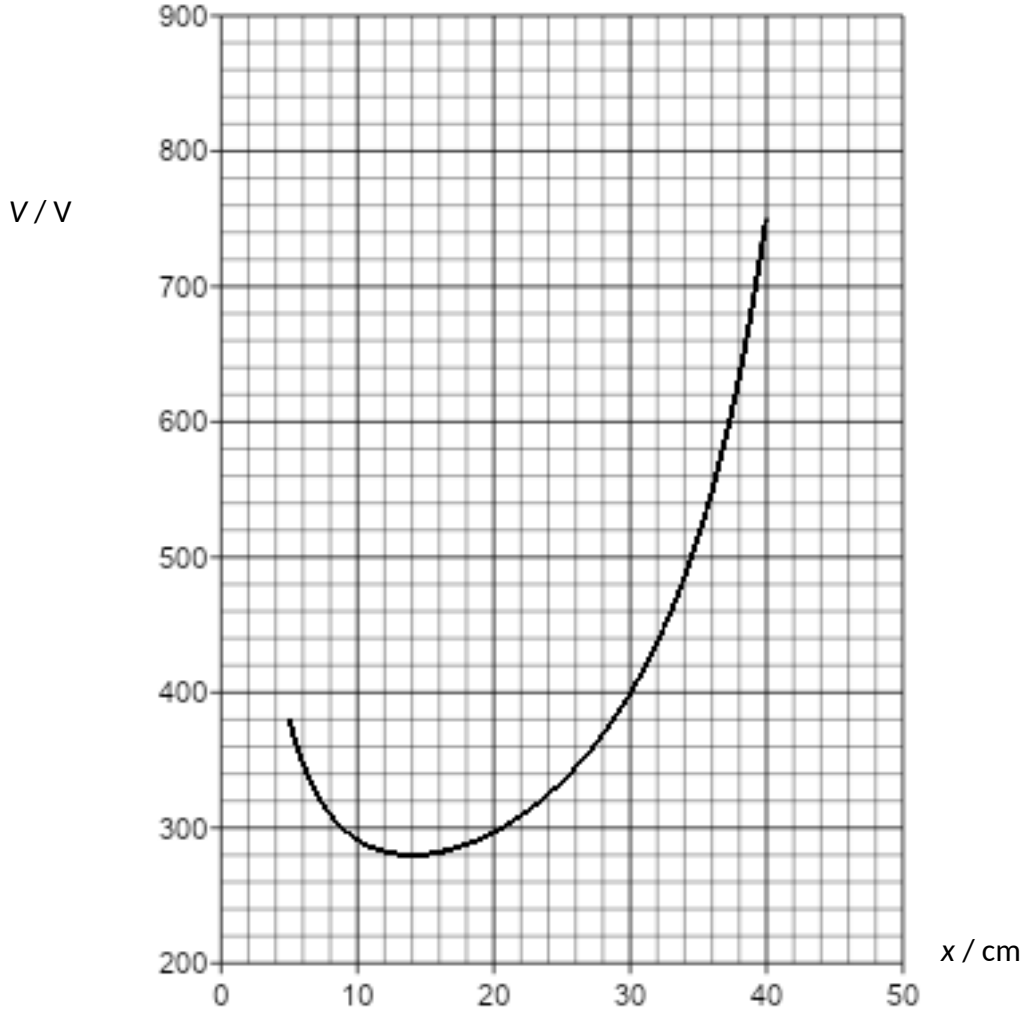


Fig. 5.2

(a) Explain why both charges are positive

……

…… [1]

(b) Describe how the resultant electric field will vary, along the line joining the centres of the spheres, from the surface of sphere A to the surface of sphere B.

……

……

……

……

…… [3]

(c) Show the electric field strength at x = 30 cm is 1650 N C^{-1}.

[1]

(d) The charge on each sphere may be considered to be a point charge at its centre. The magnitude of charge of sphere B is 8.0 nC.

Using the information in **(c)**, determine the magnitude of the charge, in nC, of sphere A.

charge of sphere A =..... nC [2]

(e) An electron is ejected from the surface of sphere B. It moves towards sphere A along the line joining the centre of the two spheres.

Determine, the minimum speed that will allow the electron to reach the surface of sphere A.

minimum speed = ... m s^{-1} [3]

[Total: 10]

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6 A rigid metal coil with 60 turns measuring 12.0 cm by 8.5 cm is suspended from a sensitive newton-meter, as shown in Fig. 6.1. The metal coil is initially above a region of uniform magnetic field of flux density *B*.

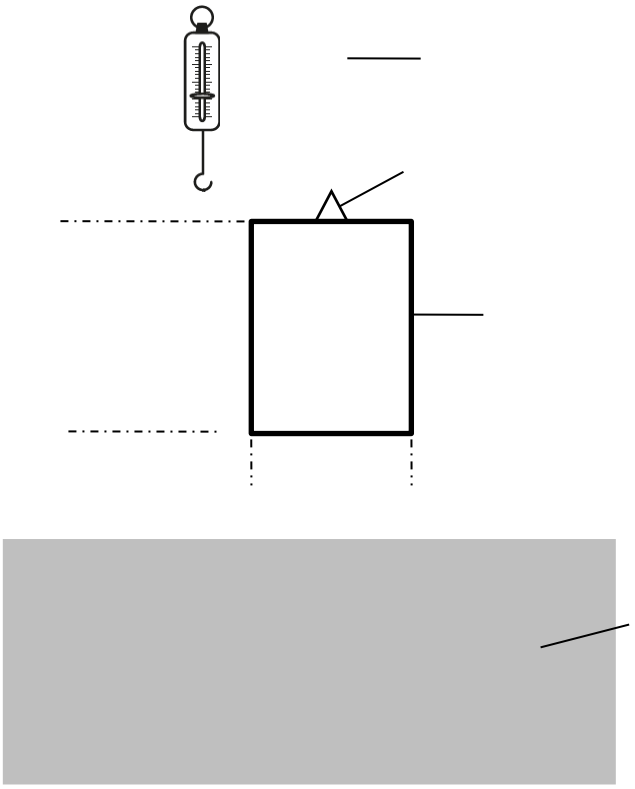


Fig. 6.1

(a) The coil is connected to a power supply so that there is a current of 6.0 A in the direction indicated in Fig. 6.1. The magnetic field is perpendicular to the page.

The coil is lowered into the uniform magnetic field until all of side PQ is just in the field. At equilibrium, the reading on the newton-meter is less than the reading when the coil is at the position shown in Fig. 6.1.

State and explain the direction of the magnetic field.

……

……

…… [2]

(b) When the side PQ of the coil is in the magnetic field in **(a)**, the newton-meter reading is 101.35 g.

The direction of the current is reversed and the current remains at 6.0 A. The newton-meter reading becomes 125.76 g at equilibrium.

Determine the magnetic flux density of the uniform magnetic field.

magnetic flux density = …… T [3]

(c) State and explain whether the reading on the newton-meter will differ from **(a)** if half of the coil is now in the magnetic field.

……

……

……

…… [2]

[Total: 7]

7 A small coil C has 64 turns and cross-sectional area 0.71 cm^2. The coil is placed inside a solenoid as shown in Fig. 7.1.

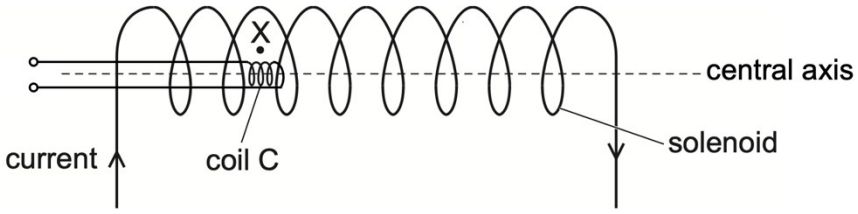


Fig. 7.1 (not to scale)

The centre of coil C is on the central axis of the solenoid.

(a) Coil C is held stationary at X. The current in the solenoid varies so that the magnetic flux density *B* at X varies from time 0 to time 4*t* as shown in Fig. 7.2.

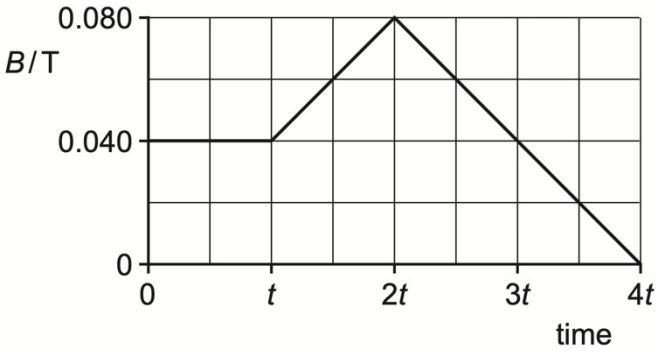


Fig. 7.2

(i) Calculate the maximum magnetic flux linkage in coil C.

flux linkage = …… Wb [2]

(ii) On Fig. 7.3, sketch a line to show the induced electromotive force (e.m.f.) *E* in coil C from time 0 to time 4*t*.

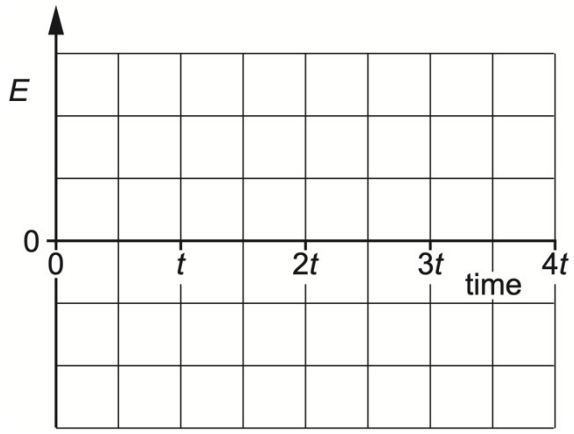


Fig. 7.3

[2]

(b) The terminals of coil C are connected together. Coil C is then removed from inside the solenoid.

With reference to the magnetic fields of the solenoid, explain why a force is required to remove coil C from the solenoid.

……

……

……

……

……

…… [3]

[Total: 7]

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H **IATIONAL JUNIOR COLLEGE**

ENIOR HIGH 2 PRELIMINARY EXAMINATION

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Section B
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For Examiner's Use	
Section B	
8	**/ 20**
9	**/ 20**
deduction	
Total (20)	

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DATA AND FORMULAE

Data

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	$\left(\quad = 8.99 \times 10^9 \text{ m F}^{-1}\right)$
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[Turn over

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electric current $I = nAvq$

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radioactive decay $x = x_0 e^{-\lambda t}$

radioactive decay constant $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **one** question from this section in the spaces provided.

8 **(a)** The kinetic theory of gases showed that the pressure p of an ideal gas is related to the mean-square speed $\langle c^2 \rangle$ of the gas molecules according to equation 8.1.

$$pV = \frac{1}{3} Nm \ \ c^2 \qquad \text{(equation 8.1)}$$

In deriving equation 8.1, the force F exerted by N molecules, each of mass m, on one wall of a cubical container of volume V is given by equation 8.2.

$$F = ¿ \left(\frac{N}{t} \right) \Delta p \qquad \text{(equation 8.2)}$$

where Δp is the change in momentum of one gas molecule after colliding with the wall and $\left(\frac{N}{t} \right)$ is the rate of collision of the molecules with the same wall.

Two of the assumptions used in deriving equation 8.1 are:

Assumption 1: The duration of collision between each molecule and the wall is negligible compared to the time spent in between collision with the same wall.

Assumption 2: Total volume of the gas molecules is negligible compared to the volume occupied by the gas.

Each gas molecule is modelled as a hard sphere.

Explain how the rate of collision and the pressure exerted by the gas will defer from the values predicted by the kinetic theory if

(i) assumption 1 becomes invalid while the rest of the assumptions remained valid,

collision rate: ……..……..

……..……..

…… [2]

gas pressure: ……

..……..

…… [2]

(ii) assumption 2 becomes invalid while the rest of the assumptions remained valid.

collision rate: ……..……...

……..

…… [2]

gas pressure: ……

……

…… [2]

(b) A heat engine is a machine that uses the heat supplied to do mechanical work. The working substance is a fixed mass of ideal gas enclosed in a cylinder with a frictionless movable piston. The gas is made to undergo the changes in pressure p and volume V shown in Fig. 8.1.

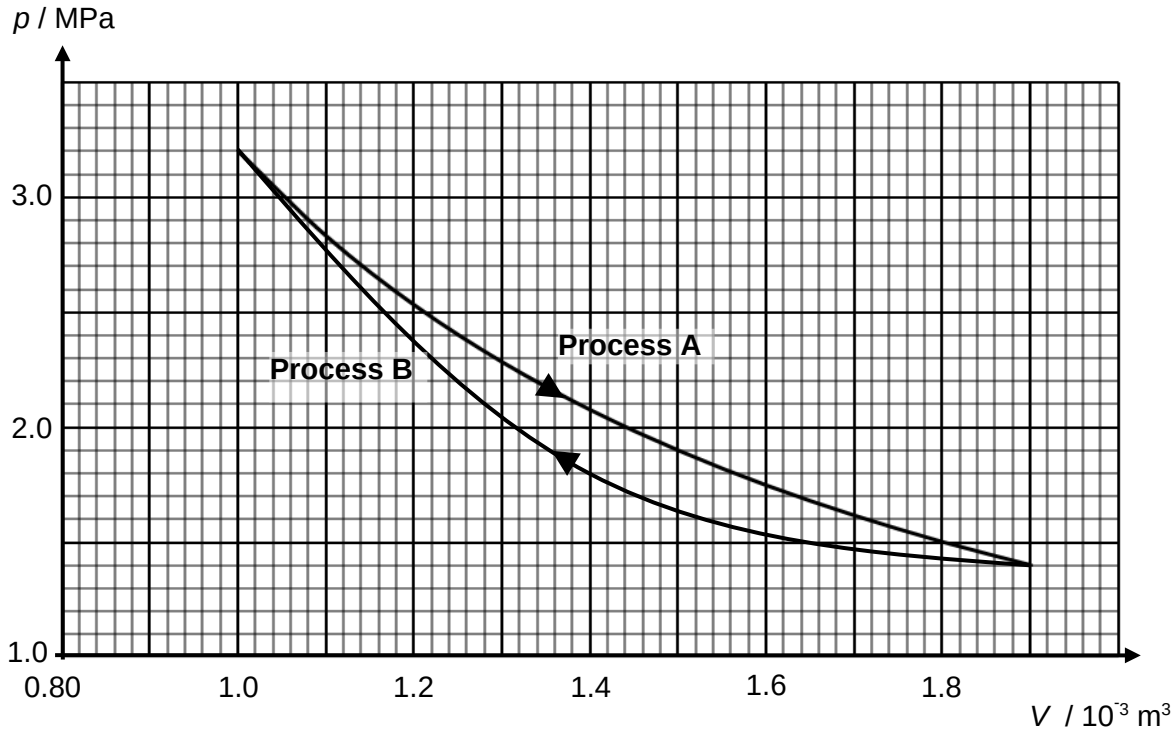


Fig. 8.1

(i) Use data from Fig. 8.1 to show quantitatively that process A cannot take place at constant temperature.

[2]

(ii) There is no transfer of thermal energy between the gas and the surrounding during process A.

Explain, using the kinetic theory of gases, why the temperature of the gas decreases during process A.

……

……

……

……..

…… [3]

(iii) The pressure and volume of the gas in process A are related according to equation 8.3.

$$pV^{\beta} = \text{¿ constant} \qquad \text{(equation 8.3)}$$

where β is a constant.

1. Use data from Fig. 8.1 to determine the value of the constant β.

β = …….. [2]

2. State the unit of β.

unit of β: …….. [1]

(iv) Estimate the work done on the gas during process B.

work done on gas = …… J [2]

(v) The efficiency of a heat engine is given by

$$\text{efficiency ¿ } \frac{net\ \ work\ \ done\ \ by\ \ gas}{heat\ \ supplied} \times 100\,\%$$

Explain why a heat engine based on the cyclic process shown in Fig. 8.1 cannot exist in practice.

……..

……..

……..
[2]

[Total: 20]

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9 **(a)** Fig. 9.1 represents part of the emission spectrum of atomic hydrogen. It contains a series of lines and the wavelength of some of which are indicated. There are no lines in the series with wavelengths less than 91.2 nm.

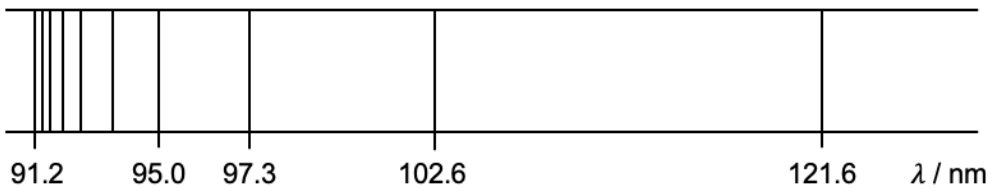


Fig. 9.1

(i) Show that the energy E of a photon, in eV, and its wavelength λ, in nm, are related by the equation:

$$E\lambda = 1243 \text{ eV nm}$$

[2]

(ii) The photon energies of the 91.2 nm, 102.6 nm and 121.6 nm lines are 13.6 eV, 12.1 eV and 10.2 eV respectively.

Determine the photon energies of the remaining two wavelengths indicated in Fig. 9.1.

photon energy of 95.0 nm = …….. eV

photon energy of 97.3 nm = …….. eV

[2]

(iii) The highest energy level for hydrogen is 0 eV.

The wavelengths in Fig. 9.1 are produced when the atom de-excite from higher energy to the ground state.

Use the values in **(a)(ii)** to construct the partial energy level diagram of the atom in Fig. 9.2.

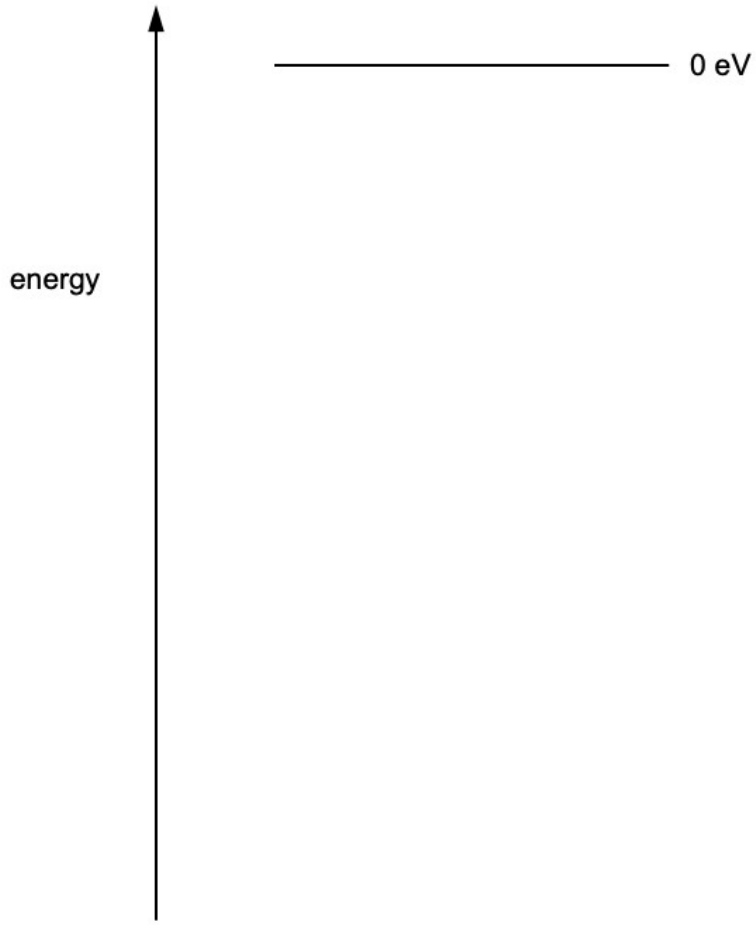


Fig. 9.2

[5]

(iv) Another line in the hydrogen occurs at a wavelength of 430 nm. Label on Fig. 9.2 the likely transition responsible for this line. [1]

(b) Fig. 9.3 shows two allowed wavefunctions ψ_A and ψ_B for an electron confined in a one-dimensional infinite square potential well of length *L*.

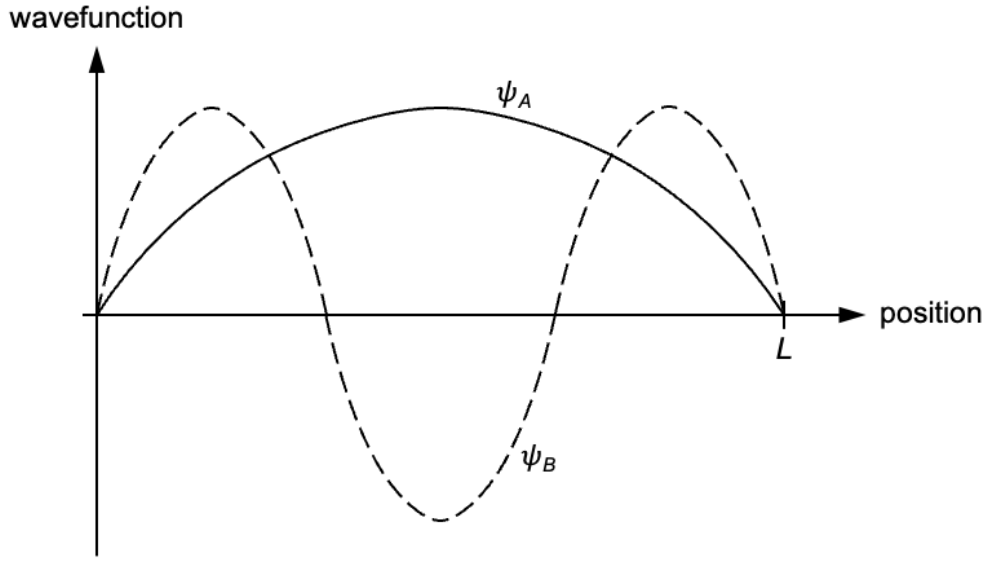


Fig. 9.3

(i) Explain which wavefunction corresponds to a higher electron energy level.

……..

……..

……..

……..

…… [3]

(ii) *L* is 1.0×10^{-10} m.

Determine the energy of the photon emitted when the electron transits from the higher energy level to the lower energy level.

energy of photon = …… J [3]

(c) A beam of electrons is directed along a normal towards a barrier, as shown in Fig. 9.4

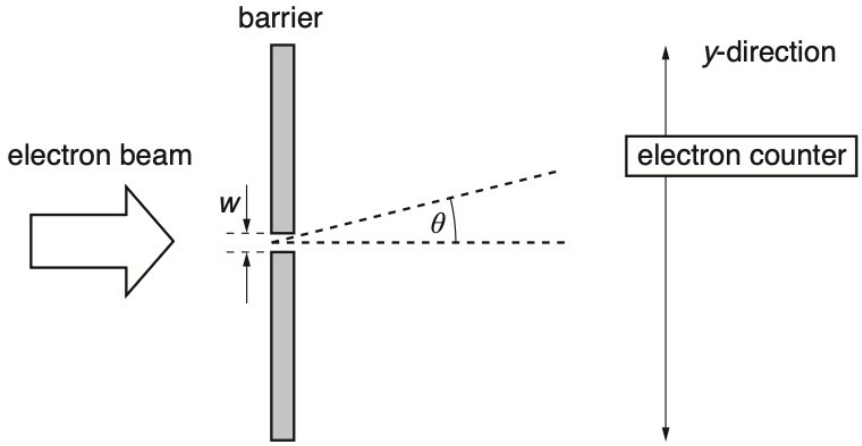


Fig. 9.4

The barrier contains a single slit of width *w*. Beyond the slit there is a detector that counts electrons. This can be moved in the *y*-direction to compare the rate of arrival of electrons at different values of the angle θ from the original direction of the beam.

Louis de Broglie suggested that some aspects of the behaviour of electrons can be explained using a wave model. The diffraction of the matter wave explains why the electron beam spreads out beyond the slit.

Werner Heisenberg used a different approach involving what is now known as the uncertainty principle. This can also be used to explain why the electron beam spreads out after passing through the slit.

(i) Calculate the uncertainty in momentum when an electron travelling at 2.8×10^{7} m s^{-1} towards the barrier passes through a narrow slit of width 1.5×10^{-10} m

uncertainty in momentum = …… N s [2]

(ii) Use the uncertainty principle to explain why the electron beam spreads out beyond the slit.

……..

……..

[Turn over

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…… [2]

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